BusFareTrack: A Public Bus Trip and Fare/Revenue Monitor

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Database Development with PL/SQL
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1. Problem Statement

Public buses carry thousands of passengers every day, and every ride produces a fare that must reach the operator. When fares are recorded in different places or checked by hand, some of the money is lost without anyone noticing. This is known as revenue leakage. Officials describing Kigali's new public transport model said that fare collection was outdated, that revenue leaked, and that there was limited visibility of how much money was actually collected (The New Times, 2025).

Rwanda has already moved a long way toward cashless fares. Tap&Go smartcards were launched in December 2015, and passengers load credit and tap the card when boarding (Africanews, 2016). Public buses became digital in 2017 (Africa Sustainability Matters, n.d.), and within three months of the roll-out the first operator reported almost 70,000 active cards, about 20% of the Kigali bus market (Consultative Group to Assist the Poor [CGAP], n.d.). Research on this system has studied whether passengers accept cashless payment (Nzaramyimana, n.d.). What is still missing is a simple, well-designed database that connects trips, buses and fares, so a manager can check revenue and notice when it falls below what is expected. The people affected are bus operators, city transport managers and the passengers who depend on well-funded, reliable buses.

2. Proposed Solution

BusFareTrack is an Oracle database with PL/SQL programs that follows a fare from the moment it is paid to the daily revenue report. A clerk or system calls a procedure to record each fare. The database checks that the bus is in service and the amount is valid before saving it. At the end of the day, another procedure adds up the fares for every route and bus and compares the result with the expected revenue for the trips that were made. When actual revenue is much lower than expected, the database creates an alert. Because these rules live inside the database, they apply no matter which tool or user enters the data, and every attempt is written to an audit log.

3. Data Source

The project uses the Salvador Unified Network for Transport (SUNT) dataset. It contains smart-card fare validations with an anonymized card number, the time of validation, the bus line, the direction, the fare paid and the vehicle, together with route, stop and trip tables in GTFS format (SUNT dataset, n.d.). A sample of at least a few hundred records will be loaded, so no personal information is stored. Before loading, the licence will be checked to confirm that educational use is allowed. The data comes from another city because Kigali fare data is not publicly available. The database design follows the smartcard model used in Kigali, so the same design could be used with local data later.

4. Planned Tables

The database will have eight tables in third normal form. OPERATORS stores the bus companies. ROUTES stores each bus line, its direction and base fare, and belongs to one operator. VEHICLES stores each bus, its status (in service or not) and its operator. TRIPS stores each journey of a vehicle on a route. FARE_TRANSACTIONS stores each fare paid on a trip, with the card hash, time and amount. DAILY_REVENUE stores the total revenue per route, vehicle and day. REVENUE_ALERTS stores warnings when revenue is below expected. AUDIT_LOG stores who tried to change data, when, and whether it was allowed. One route has many trips, and one trip has many fare transactions.

5. Planned PL/SQL Features

Procedures: record_fare validates a fare and inserts it into FARE_TRANSACTIONS; close_day builds the DAILY_REVENUE rows and locks that day.

Functions: get_route_revenue returns the revenue of a route on a date; get_expected_revenue estimates what a trip should earn from its route and past fares.

Cursors: a cursor loops through the day's trips, compares actual with expected revenue and inserts a row into REVENUE_ALERTS when the difference is large.

Package: bus_revenue_pkg groups the procedures and functions so the dashboard and users call one clean interface.

Triggers: one trigger blocks a fare on a vehicle that is not in service; another blocks any change to fares of a closed day. Both write the attempt to AUDIT_LOG.

Exceptions: custom errors for an unknown vehicle, a closed day and an invalid amount give clear messages.

6. Innovation

The innovation is an analytics dashboard built with Oracle APEX and connected live to the Oracle database. It shows revenue by route and day, the number of trips per bus and the open revenue alerts, so a manager can see problems without writing SQL. The dashboard only displays data and calls the PL/SQL procedures; the business rules stay in the database. It will be demonstrated live in Phase VII.

References

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